



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2024-25/EMDLAB /591816

Date:

16/10/2024(18:04:13)

Subject:REVISION OF LMIOINOPSELEC044 INSPECTION OF
BRUSHLESS EXCITER – BHEL make

Subject: REVISION OF LMI/OIN/OPS/ELEC/044 INSPECTION OF BRUSHLESS EXCITER
– BHEL make

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes envisaged by COS. Only periodic reviewing done. No changes in the document.

This LMI has been prepared in line with new LMI format.

May please approve.

Submitted by:

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On:16/10/2024(18:04:13) **Ref:**SMTTP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:2

LMI reviewed in 2023, revised inline with standard LMI format.

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On:16/10/2024(18:13:02) **Ref:**SMTPP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:3

Periodic review

Submitted by:

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Note No:4

formatting is done as per New format,It should be Rev.02 as Rev.01 is superceded

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On:17/10/2024(10:17:15) **Ref:**SMTTP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:5

Updated.

Submitted by:

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On:22/10/2024(09:56:12) **Ref:**SMTTP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:6

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On:22/10/2024(09:59:42) **Ref:**SMTTP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:7

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On:22/10/2024(16:25:37) **Ref:**SMTPP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:8

Updated

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On:22/10/2024(18:18:54) **Ref:**SMTPP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:9

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On:23/10/2024(09:37:59) **Ref:**SMTPP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:10

LMI is in line with Operation Information Note.

Submitted by:

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On:25/10/2024(18:57:15) **Ref:**SMTTP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:11

Reviewed and submitted for approval please

Submitted by:

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On:28/10/2024(16:29:17) **Ref:**SMTTP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:12

May please be approved

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On:28/10/2024(19:18:58) **Ref:**SMTTPP/ELECT MAINT/2024-25/EMDLAB /591816

Note No:13

Approved & Submitted by:

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

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On:28/10/2024(19:31:35) **Ref:**SMTTPP/ELECT MAINT/2024-25/EMDLAB /591816

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

INSPECTION OF BRUSHLESS EXCITER

-BHEL MAKE

LMI/OIN/OPS/ELEC/044 Rev:02

OCT 2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	INSPECTION OF BRUSHLESS EXCITER – BHEL MAKE
2	LMI No.	LMI/OIN/OPS/ELEC/044 Rev:02
3	Reference Document	OIN/OPS/ELEC/044
4	Superseded Document	LMI/OIN/OPS/ELEC/044 Rev:01 OCT 2023
5	Prepared By	Name: Ramani Kanta Jena S M (EMD)
6	Checked By	Name: Balabalaji Akula DGM (EMD)
7	Reviewed By	Name: T Vijay Krishna A G M (EM)-I/C
8	Reviewed By	Name: Om Prakash A G M (Maint)
9	Reviewed By	Name: P K Jena G M (O&M)
	Approved By	Name: S.K.Sinha H O P (Simhadri)

TITLE	INSPECTION OF BRUSHLESS EXCITER -BHEL MAKE
LMI No.	LMI/OIN/OPS/ELEC/044 Rev:02

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TITLE	INSPECTION OF BRUSHLESS EXCITER -BHEL MAKE
LMI No.	LMI/OIN/OPS/ELEC/044 Rev:02

1.0 Introduction

This LMI provides guidance on scope of inspection with schedule and recommends various activities to be carried out during service / opportunity shutdown to eliminate the chance of entanglement of stroboscope cables / instrumentation cables with rotating parts and to find out any looseness of resi-glass bandage used for securing AC lead from armature to diode wheel.

2.0 Superseded Documents

LMI/OIN/OPS/ELEC/044 Rev:01 OCT 2023

3.0 Details of Revision

NIL

4.0 METHODOLOGY OF IMPLEMENTATION

The following inspections are mandatory and have to be carried out as per the schedule mentioned in the table below.

Sr No	Description of job	Schedule
1	Stroboscope light & reflector assembly & viewglasses to be checked for proper & secure fixing.	Opportunity shutdown / Once in six months, whichever is later
2	Cables used for stroboscope lights to be kept away from rotating parts and to be dressed & secured. No cable loop to be kept near diode wheel cover or inside inner chamber of exciter. Fixing of cable to be checked for any looseness and subsequent rectification.	Opportunity shutdown / Once in six months, whichever is later

TITLE	INSPECTION OF BRUSHLESS EXCITER -BHEL MAKE
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3	Cables used for bearing vibration & temperature monitoring to be kept away from rotating parts and to be dressed & secured. No cable loop to be kept near diode wheel cover or inside inner chamber of exciter. Fixing of cable to be checked for any looseness and subsequent rectification.	Opportunity shutdown / Once in six months, whichever is later
4	Inspection of various installations i.e. lighting fixtures, bearing chamber windows, rotor earth fault monitoring brush holder fixing arrangement, air guides for proper fixing / defect & rectification.	Opportunity shutdown / Once in six months, whichever is later
5	Inspection of various installations i.e. RTDs, temperature switches, vibration sensors fixing arrangement / defect & rectification.	Opportunity shutdown / Once in six months, whichever is later
6	Visual inspection of PMG, Main exciter field and armature, diode wheel, Resi-glass bandage & Textolite half ring, spacers, AC copper leads etc. to detect any damage / shifting / looseness.	Overhauling
7	Inspection of all welding, nuts & bolts and their locking arrangement for any defect and rectification.	Overhauling
8	Inspection and cleaning of air filters	Opportunity shutdown / Once in six months, whichever is later

TITLE	INSPECTION OF BRUSHLESS EXCITER -BHEL MAKE
LMI No.	LMI/OIN/OPS/ELEC/044 Rev:02

5.0 FREQUENCY OF CHECKING AND MEASUREMENTS

During opportunity shutdown / once in six months, whichever is later / overhauling.

6.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION

Basic responsibility lies with respective Area in-charge of EM and C&I (for Methodology of implementation Sl. No.3 &5). Overall responsibility lies with AGM (EM).

7.0 EXCEPTION REPORTING & DISPENSATION OF DEVIATION

Any deviation shall be reported to HOD (EM) and can be dispensed with his approval.

8.0 REVIEW

Documents will be reviewed once in two years by HOP.

9.0 CHECKLIST

TITLE	INSPECTION OF BRUSHLESS EXCITER -BHEL MAKE
LMI No.	LMI/OIN/OPS/ELEC/044 Rev:02

CHECKLIST FOR INSPECTION OF EXCITER

(OPPORTUNITY SHUTDOWN / ONCE IN 6 MONTHS WHICHEVER IS LATER)

UNIT:

DATE:

SL No	Description of job	Observations	REMARKS
1	Stroboscope light & reflector assembly & view glasses to be checked for proper & secure fixing.		
2	Cables used for stroboscope lights to be kept away from rotating parts and to be dressed & secured. No cable loop to be kept near diode wheel cover or inside inner chamber of exciter. Fixing of cable to be checked for any looseness and subsequent rectification.		
3	Cables used for bearing vibration & temperature monitoring to be kept away from rotating parts and to be dressed & secured. No cable loop to be kept near diode wheel cover or inside inner chamber of exciter. Fixing of cable to be checked for any looseness and subsequent rectification.		
4	Inspection of various installations i.e. lighting fixtures, bearing chamber windows, rotor earth fault monitoring brush holder fixing arrangement, air guides for proper fixing / defect & rectification		
5	Inspection of various installations i.e. RTDs, temperature switches, vibration sensors fixing arrangement / defect & rectification.		
6	Inspection and cleaning of air filters		

TITLE	INSPECTION OF BRUSHLESS EXCITER -BHEL MAKE
LMI No.	LMI/OIN/OPS/ELEC/044 Rev:02

CHECKLIST FOR INSPECTION OF EXCITER-OVERHAULING

UNIT:

DATE:

SL no	Description	Observations	Remarks
1	Visual inspection of PMG, Main exciter field and armature, diode wheel, Resi-glass bandage & Textolite half ring, spacers, AC copper leads etc. to detect any damage / shifting / looseness.		
2	Inspection of all welding, nuts & bolts and their locking arrangement for any defect and rectification.		